

Classical DSP + GBM vs Deep Learning: A Measured Decision Map for Signal Tasks

Where classical methods win and where deep learning wins, from measurements

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Abstract

A widespread assumption in the wireless signal processing community holds that deep learning eventually beats classical digital signal processing (DSP) and gradient-boosted machines (GBM) on every task. This paper narrows that assumption into testable claims and measures them. Across six tasks - modulation classification, fixed-false-alarm-rate interference detection, drone open-set rejection, channel estimation, RF fingerprint fusion, and co-channel signal separation - classical methods and deep learning were compared head-to-head on the same data, the same splits, and the same metrics. The results split by task rather than settling on a single verdict. On the standard public benchmark RadioML2016.10a (6dB SNR subset), classical methods (28-dimensional physical features plus LightGBM, accuracy 0.855) clearly beat deep learning (a 1.39M-parameter transformer, 0.681), and this advantage did not reverse as training data grew from 539 to 9,900 samples (the gap narrows, but no crossover was observed within the measured range). For fixed-false-alarm-rate interference detection, classical CFAR (constant-false-alarm-rate) detectors held their target false-alarm rate structurally even under a shifting noise floor, while an unconstrained binary deep-learning detector saw its realized false-alarm rate explode to 1.0 under the same drift. On the public dataset DroneRF (CC BY 4.0), for drone RF open-set rejection at a single observation ($K=1$), classical feature-plus-Mahalanobis rejection achieved 5.7 times the unknown-device rejection rate of a deep-learning baseline. Conversely, deep learning won clearly on other tasks. For separating two overlapping co-channel signals, a learned separator (a Conv-TasNet variant) achieved SI-SDRi of +3.01dB, while classical independent component a

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1 Abstract

A widespread assumption in the wireless signal processing community holds that deep learning eventually beats classical digital signal processing (DSP) and gradient-boosted machines (GBM) on every task. This paper narrows that assumption into testable claims and measures them. Across six tasks - modulation classification, fixed-false-alarm-rate interference detection, drone open-set rejection, channel estimation, RF fingerprint fusion, and co-channel signal separation - classical methods and deep learning were compared head-to-head on the same data, the same splits, and the same metrics. The results split by task rather than settling on a single verdict. On the standard public benchmark RadioML2016.10a (6dB SNR subset), classical methods (28-dimensional physical features plus LightGBM, accuracy 0.855) clearly beat deep learning (a 1.39M-parameter transformer, 0.681), and this advantage did not reverse as training data grew from 539 to 9,900 samples (the gap narrows, but no crossover was observed within the measured range). For fixed-false-alarm-rate interference detection, classical CFAR (constant-false-alarm-rate) detectors held their target false-alarm rate structurally even under a shifting noise floor, while an unconstrained binary deep-learning detector saw its realized false-alarm rate explode to 1.0 under the same drift. On the public dataset DroneRF (CC BY 4.0), for drone RF open-set rejection at a single observation ($K=1$), classical feature-plus-Mahalanobis rejection achieved 5.7 times the unknown-device rejection rate of a deep-learning baseline. Conversely, deep learning won clearly on other tasks. For separating two overlapping co-channel signals, a learned separator (a Conv-TasNet variant) achieved SI-SDRi of +3.01dB, while classical independent component analysis (ICA) scored -14.25dB - worse than the unseparated mixture itself. Deep learning also beat its own classical fusion baseline on RF fingerprint fusion (AUROC 0.9994 vs 0.945 at $K=50$ fusion), but that win carries several layers of caveats. Channel estimation remains an open question: an early experiment with a methodology flaw produced only an ambiguous signal, and no confirmed verdict exists yet. This paper’s goal is not to cheer for either side, but to publish, task by task, what actually wins - together with the limits of those numbers.

2 Background

Deep learning has been treated as the default choice in wireless signal processing for several years now. Whether the task is modulation classification, interference detection, or RF fingerprint recognition, a large share of published papers and vendor materials show deep learning outperforming classical methods. Many of these comparisons, however, are not fair. The deep-learning side is often given large training sets and GPU tuning time, while the classical side is represented by untuned shallow features or a simple threshold. The opposite error is also common: leaving deep learning undertrained, or comparing only under low-data conditions that favor classical methods. Neither direction gives an honest answer to the question of who actually wins.

The question this paper asks is narrow and specific: **on the exact same data, the exact same split, and the exact same metric, which of classical DSP+GBM or deep learning actually wins on each specific wireless signal processing sub-task?** That this cannot be answered with a single sentence - “deep learning is better” or “classical is better” - is the first, and most important, finding of this research. The winner differed task by task, and so did the reason. Some tasks had training data too small for deep learning to use its capacity; some were deterministic physical problems where a learned representation was never necessary in the first place; and some involved subtle hardware signatures that hand-designed features could not easily capture, which is exactly where deep learning was needed.

The value of this paper lies less in any single verdict than in the measurement discipline kept to reach it. Every number comes from an experiment that was actually run; expected results of experiments that have not yet executed are never cited as results. Numbers from synthetic data and numbers from real public datasets are always labeled separately. Early experiments with methodology flaws are recorded together with those flaws, not quietly corrected away. This is the principle that runs through the entire paper.

3 Evaluation Methodology

Six principles were kept across all six tasks.

Same data, same split, same metric. Classical methods and deep learning share the same training/evaluation split, the same input window size, and the same evaluation metric. No preprocessing or

augmentation advantage was given to either side.

Fixed-false-alarm-rate discipline. For detection-shaped tasks, a “detection rate of X%” number is never used as the final result on its own, because lowering the threshold lets anyone raise detection rate, at the cost of false alarms. Thresholds are inverted from interference-free training-split statistics to hit a target false-alarm rate, and that threshold is applied unchanged to the test split - the standard CFAR (constant-false-alarm-rate) discipline from radar signal processing.

Oracle vs. naive baseline separation. This is a real problem encountered in the channel-estimation experiment: some cases where deep learning appeared to “win” turned out to be against a weaker classical baseline (naive least-squares, LS) rather than the stronger one (an oracle-grade MMSE that knows the true channel statistics). Once this was recognized, every subsequent comparison keeps a full classical performance ladder (from naive to oracle-grade) and states explicitly which baseline the win is measured against.

Explicit synthetic vs. real-data labeling. Measuring real wireless signals requires real RF hardware and capture equipment, and not every task in this paper has such a real capture available. Results from synthetic data are not meaningless, but they are not cited with the same weight as results from real data. Each section states its data provenance.

Unfinished GPU runs are not cited as results. Even when code and experiment design are ready, if no completed result artifact exists, the task is marked “unresolved” rather than described as a confirmed number.

4 Results by Task

4.1 Modulation Classification - Classical Wins

Classifying the modulation scheme of a radio signal (BPSK, QPSK, 16QAM, etc.) from the signal itself. On the standard public benchmark **RadioML2016.10a**’s 6dB SNR subset (11 modulation classes, 128-sample IQ window, a fixed split of 539 training / 1,100 test examples), classical methods (28-dimensional physical features - instantaneous amplitude/phase, carrier frequency offset, Swami-Sadler higher-order cumulants, and more - plus a tree-ensemble classifier) were compared against deep learning (a 1.39M-parameter mixture-of-experts transformer on raw IQ input, trained to convergence over 600 epochs) on the identical train/test split.

Method	Accuracy (11-way)
LDA (28 physical features)	0.668
RandomForest	0.840
XGBoost	0.848
LightGBM	0.855
CatBoost	0.848
Deep learning (transformer, 600 epochs, converged)	0.681
Chance level	0.091

The same 28 features were also tested against recent modern tabular learning models. TabPFN v2 came close at 0.848 but did not beat LightGBM, and deep tabular models (TabM 0.781, RealMLP 0.778) and a stacking ensemble (AutoGluon 0.818) also fell short of the gradient-boosted tree ensemble. This suggests the answer is not “a better tabular model” - at this data scale, gradient-boosted tree ensembles are already close to saturation.

Does scaling up the data reverse this? If deep learning’s disadvantage was simply a lack of data, the hypothesis was that scaling up training data would reverse the verdict. This was tested separately, re-measuring on the same 6dB data while only increasing training sample count.

Training samples	Classical (LightGBM)	Deep learning (transformer)	Gap (DL - classical)
539	0.728	0.368	-0.360
2,000	0.794	0.581	-0.213

Training samples	Classical (LightGBM)	Deep learning (transformer)	Gap (DL - classical)
5,000	0.814	not measured	-
9,900	0.844	not measured	-

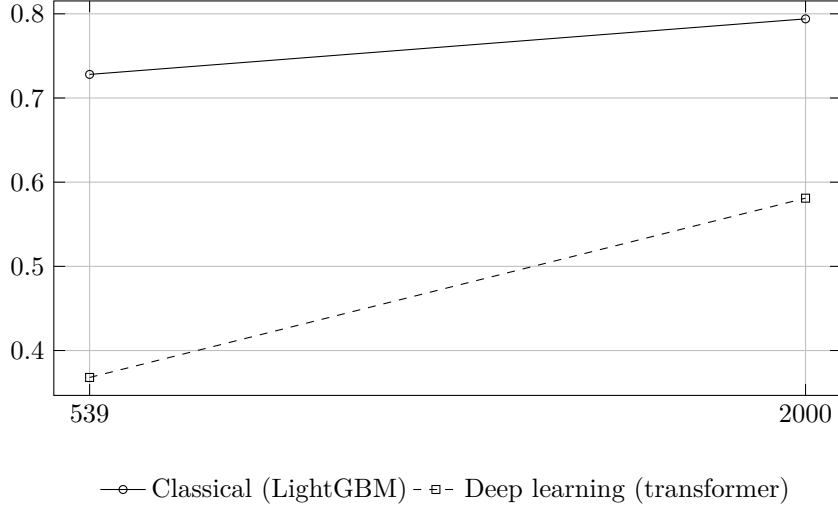


Figure 1: Accuracy vs training set size

Classical led across every measured point, but deep learning closed the gap quickly as data grew (gap -0.360 $\rightarrow$ -0.213). The classical curve plateaued gently around 0.84, and the deep-learning re-measurement at 5,000 and 9,900 samples was not completed in this run, so whether a crossover actually exists was not observed within this data range (up to 9,900, a single SNR). To be transparent, this table’s classical accuracy at $N=539$ (0.728) differs from the same $N=539$ point in the table above (0.855). The two experiments use the same underlying data but different subsampling procedures, and since the purpose of this scaling experiment is the relative slope of the two curves rather than absolute accuracy, this discrepancy does not affect the conclusion.

Separately, whether deep-learning pretraining (unsupervised, via masked IQ-segment reconstruction) helps under label scarcity was also checked. At 1% labels (99 examples), random initialization scored 0.212 vs. 0.273 (+0.062) with pretraining; at 10% labels (990 examples), 0.504 vs. 0.554 (+0.050). The gain was consistent but modest. At full labels (9,900), random initialization was slightly ahead (0.797 vs. 0.787). This low-label local benefit is real, but it does not reverse the overall verdict of this section.

4.2 Fixed-False-Alarm-Rate Interference Detection - Classical Has a Structural Advantage

Detecting wireless interference (jamming, unauthorized transmission, carrier frequency offset, etc.) around a factory floor or medical device is a task where the real contest is not the accuracy number but **whether the false-alarm rate is actually held fixed**. This was measured on two synthetic benches: an O-RAN/5G-style multi-slice scenario (concept-level, not a standards-compliance test) and a second bench that deliberately injects noise-floor drift to test whether the false-alarm-rate guarantee survives a distribution shift.

On the first bench, a cell-averaging CFAR (CA-CFAR) detector combined with a carrier-frequency-offset estimator achieved the following detection rates at a target false-alarm rate of 5% (realized 5.29%):

Impairment	Detection rate (Pd)	Slice localization accuracy
Barrage (full-band) jamming	0.929	0.929
Partial-band jamming (25%)	0.986	0.991

Impairment	Detection rate (Pd)	Slice localization accuracy
Single-tone jamming	1.000	1.000
Carrier frequency offset only	1.000	1.000
Overall	0.980	0.982 (random baseline 0.167)

No deep-learning counterpart exists for this bench, so this is a classical-only number. Interestingly, barrage jamming - which injects the most total power - had the lowest detection rate (0.929). This is the well-known CFAR “masking” effect: a full-band jammer raises the reference-cell noise estimate almost as much as the cell under test, compressing the power-ratio statistic.

The second bench shows where the real contest lies. Several CFAR variants (CA/OS/GO/SO-CFAR), an unconstrained binary deep-learning detector, and a learned detector with CFAR discipline explicitly built into training (a “constrained” deep learning variant) were re-evaluated at severity levels held out from training, with an additional 12dB noise-floor drift injected to see whether the target false-alarm rate actually held.

Method	Realized false-alarm rate after drift, target 0.01	Realized false-alarm rate after drift, target 0.001
CA-CFAR	0.0	0.0
OS-CFAR	0.0	0.0
GO-CFAR	0.0	0.0
SO-CFAR	0.0	0.0
Cyclostationary detector	0.100	0.007
Unconstrained binary deep-learning detector	1.000	1.000
CFAR-constrained deep learning	0.040	0.003

This table shows exactly where classical methods hold a real advantage on this task. The CA/OS/GO/SO-CFAR family kept the target false-alarm rate even under a 12dB noise-floor shift - meaning the property CFAR’s name promises (a constant false-alarm rate) held structurally, with no recalibration. The unconstrained binary deep-learning detector, in contrast, saw its 0.01-target threshold blow up to a realized false-alarm rate of 1.0 under the same drift, meaning it flagged every window as interference - a complete collapse of the false-alarm guarantee. The deep-learning variant with CFAR discipline built into its loss landed in between (0.04, 0.003). In other words, “fixed false-alarm rate” is not a property that a single accuracy number can summarize - it must be measured separately under distribution shift - and on this measurement, pure classical CFAR provided the strongest guarantee.

4.3 Drone RF Open-Set Rejection - Classical Wins at Low K

Detecting and identifying drone control signals, and rejecting a new, never-seen drone (an unknown device) at inference time. Using the public dataset **DroneRF** (Al-Sa’d et al., Qatar University, Mendeley DOI 10.17632/f4c2b4n755, **CC BY 4.0**, citable commercially), a classical pipeline (24-dimensional physical features: RSSI, envelope statistics, carrier frequency offset, higher-order cumulants, spectral shape, plus LightGBM and Mahalanobis-distance open-set rejection) was reimplemented and compared, at matching fusion count K, against a same-genre deep-learning experiment (ECAPA-TDNN embeddings with contrastive learning, on a different dataset, CardRF).

K (fused observations)	Classical unknown-rejection rate	Classical AUROC	DL unknown-rejection rate	DL AUROC
1	0.424	0.648	0.074	0.565
5	0.581	0.720	0.129	0.708
10	0.725	0.811	0.408	0.761

K (fused observations)	Classical unknown-rejection rate	Classical AUROC	DL unknown-rejection rate	DL AUROC
20	0.875	0.937	0.917	0.829
50	1.000	1.000	1.000	0.914

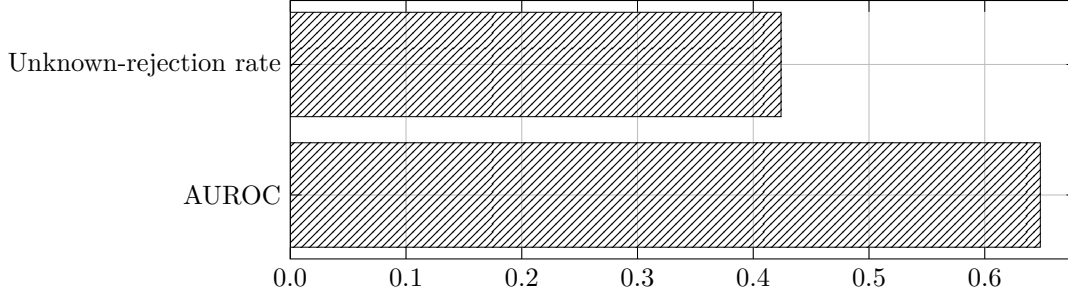


Figure 2: Drone open-set rejection at K=1

At a single observation (K=1, no fusion), classical unknown-device rejection was 5.7 times that of deep learning. The deep-learning embedding essentially extracted no usable signal from a single packet (its contrastive-learning objective had collapsed), while the 24-dimensional hand-designed features were already meaningful at a single observation. Classical led through K=5 and K=10 as well; at K=20 deep learning edged ahead slightly (0.917 vs. 0.875), and at K=50 both effectively saturated (1.0/1.0). The K=20 and K=50 points, however, involved only 40 and 16 unknown samples respectively, so this “tie” should be read as a sample-size limit rather than evidence of method superiority.

An honest comparison limit must be stated. This is not a direct A/B on the same dataset. The classical experiment used DroneRF (3 drone types, 2 known classes / 1 unknown), while the deep-learning experiment used CardRF (6 controllers, 4 known / 2 unknown, including 4 units from the same manufacturer). The task genre (drone RF open-set rejection, the same Mahalanobis rejection recipe) matches, but the data and class structure differ, so “classical beat deep learning” should be read as a same-genre comparison, not a direct head-to-head on identical data. This classical pipeline was also not strong at fine-grained manufacturer discrimination (3-class accuracy 0.5685, majority-class baseline 0.444) - it is strong at presence/absence detection and open-set rejection, but this 24-dimensional feature set was insufficient for fine manufacturer discrimination.

4.4 Channel Estimation/Equalization - Not Yet Resolved

Estimating and reversing the effect of the wireless channel on a received signal. This is the point where this paper must most honestly say “we do not know yet.” An early experiment conflated two distinct tasks - channel estimation (NMSE metric) and full-receiver performance (BER metric) - into a single verdict, and a later review found that some cases counted as deep learning “wins” were in fact wins against a weaker naive least-squares (LS) baseline rather than the stronger oracle-grade MMSE baseline. The flawed-methodology aggregate result was 37 classical wins, 11 deep-learning wins, and 51 ties out of 99 comparison cells, but this number is not cited as a conclusion.

A corrected redesign - fully separating the estimation task from the receiver task, adding a full classical performance ladder from naive LS to oracle-grade, and adding an 8-axis out-of-distribution matrix - has been prepared in code, but as of this paper’s writing, the actual GPU run has not been completed. Channel estimation is therefore left **unresolved** between classical and deep learning. A directional guess is possible (that classical/oracle-grade MMSE will likely remain strong, which was the premise motivating the corrected redesign), but this paper’s principle is not to describe a guess as a confirmed result.

4.5 RF Fingerprint Fusion - Deep Learning Wins, With Several Layers of Caveats

Authenticating a device from the subtle analog imperfections its own hardware imprints on a signal (an RF fingerprint). On the public research dataset **WiSig** (CC BY-NC-SA, non-commercial research use only), classical features (unsupervised, no GPU required, explainable) were compared against a learned embedding (an ECAPA-TDNN-family model, requiring GPU training) using multi-packet fusion (K fused packets per authentication decision).

K (fused packets)	Classical AUROC	DL AUROC
1	0.549	0.696-0.739
10	0.766	not measured
20	0.863	not measured
50	0.945	0.9994
100	0.979	not measured

At K=50 fusion, deep learning beat classical’s own baseline (0.9994 vs. 0.945). This win, however, carries at least three layers of caveats.

First, this win is against our own K=50 classical baseline, not against an external single-decision literature reference. At the only point directly comparable to published literature - a single packet, K=1, no fusion - this deep-learning embedding scored AUROC 0.696-0.739, well below a published reference value (a strict cross-receiver open-set AUROC of 0.9692).

Second, this K=50 number raised a methodology concern in a post-hoc audit. The 50 observations used in fusion may not be 50 independent authentication attempts, but correlated, consecutive packets from the same capture session - a possibility the data structure does not rule out. If so, 0.9994 would be the ceiling of a much easier task - a single-session match boosted by accumulated signal-to-noise ratio - rather than 50 independent authentication decisions.

Third, both methods are weak at the strict operating point that matters for real access control. On a real security metric - detection rate at a 0.1% false-accept rate, TAR@FAR=1e-3 - the own-protocol figures were 0.16% at K=1 and only 43% at K=100. Under a stricter session-authentication product-form condition (separating enrollment/verification by unseen device and unseen day), the K=50 TAR@FAR=1e-3 was 22.35% - about a quarter of the K=50 figure cited above (87.65%). Generalization to a commercially unrestricted independent dataset (INRIA, CC BY) still showed decent separability at K=1 (AUROC 0.868), but the same strict access-control metric (TAR@FAR=1e-3) was only 2.9%.

Taken together, “deep learning beat classical at RF fingerprinting” is true only at the benchmark level (K=50 measured the same way against K=50), not against an external single-decision SOTA reference or a real security operating point. Benchmark victory and product viability are different questions, and this is the most heavily caveated of deep learning’s wins.

4.6 Co-Channel Signal Separation - Deep Learning Wins Clearly

Separating two wireless signals whose frequencies nearly overlap in a single received mixture. Because the license of a real multi-emitter dataset (RFSS) for this task was unclear, and using it for training was conservatively judged inappropriate, the same problem structure - two linearly modulated signals overlapping at a small carrier-frequency offset, swept across signal-to-interference ratio - was self-synthesized instead. The learned separator adapts a speech-separation architecture, Conv-TasNet, to complex IQ input (a 1D encoder, a dilated-TCN separator, masks, a decoder, permutation-invariant training), and the classical baseline treats a single receiver’s two real I/Q channels as two observations for FastICA (independent component analysis) plus a Hilbert transform.

The completed GPU result was unambiguous:

Method	SI-SDRi (dB, higher is better)
Classical (FastICA + Hilbert transform)	-14.25
Deep learning (Conv-TasNet variant)	+3.01

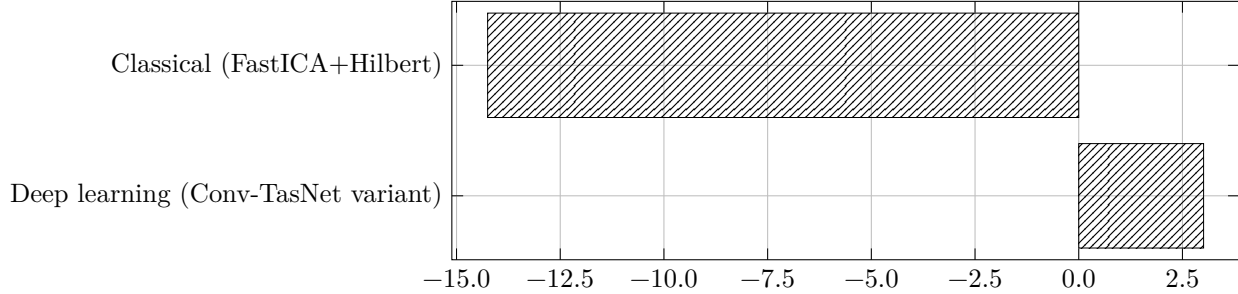


Figure 3: Co-channel RF signal separation: SI-SDRi

The classical method left the separated signal worse than the mixture itself (negative SI-SDRi). Treating a single receiver’s I/Q channels as two independent observations does not structurally satisfy the requirements of two-source independent component analysis - an expected weakness, not an experimental error. The deep-learning separator showed a clear improvement over the mixture (+3.01dB). Of the six tasks in this paper, this is the cleanest, least-caveated win for deep learning. That said, this result too comes from a self-synthesized mixture with the same structure, not from a real public multi-emitter dataset.

5 The Combined Decision Map

Summarizing the six tasks on one page:

Task	Data	Winner	Notes
Modulation classification	Real, public (RadioML2016.10a)	Classical	Holds up to 9,900 training samples
Fixed-FAR interference detection	Synthetic	Classical	Guarantee holds under drift
Drone open-set rejection (low K)	Real, public (DroneRF/CardRF, genre comparison)	Classical	DL edges ahead at $K \geq 20$
Channel estimation/equalization	Mixed real+synthetic	Unresolved	Only a flawed early result exists
RF fingerprint fusion	Real, public (WiSig, non-commercial)	Deep learning (heavily caveated)	Reverses against external SOTA / real security metric
Co-channel signal separation	Synthetic (self-generated)	Deep learning	Least-caveated DL win

What this map should convey is not a scoreboard of “classical 3, deep learning 2, one unresolved.” What decided each win or loss was the **structure** of the task. Where training data was small (hundreds to thousands of samples) and physically interpretable features existed (modulation classification, interference detection, drone open-set rejection), classical methods had the advantage. Where the task itself required a learned representation rather than a deterministic physical model - and data was plentiful (co-channel signal separation, the subtle hardware signature behind RF fingerprinting) - deep learning had the advantage. Channel estimation, where a fair experiment has not yet finished, is honestly left unresolved. The claim that “deep learning is replacing classical methods across signal processing” is not supported by the measured evidence from these six tasks.

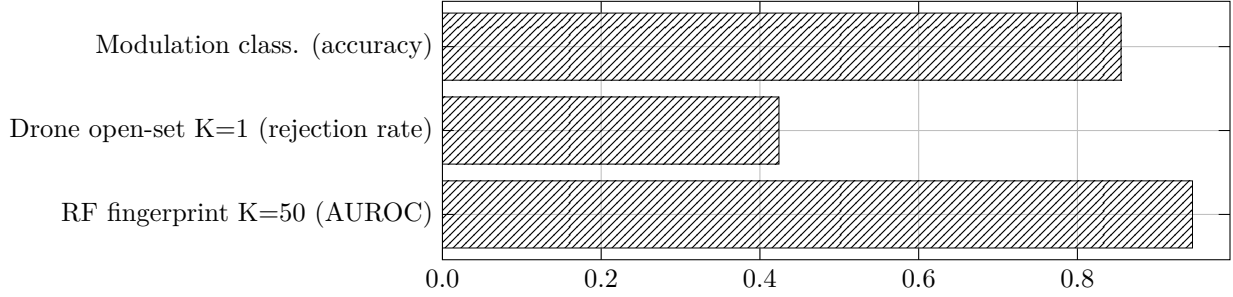


Figure 4: Best performance per task: classical vs deep learning

6 Limitations

Several limitations must be stated explicitly.

1. **Interference detection, interference localization, and the O-RAN/5G scenario are entirely synthetic data.** Real-world multipath, correlated shadowing, and real standards-compliant protocol integration are not reflected. These numbers should be read at the concept-feasibility level.
2. **Co-channel signal separation used a self-synthesized mixture with the same problem structure, not a real public multi-emitter dataset.** The real dataset’s license was unclear, so it was conservatively excluded from training use.
3. **The core dataset behind RF fingerprint fusion (WiSig) is licensed for non-commercial research use only.** These numbers are valid as benchmark evidence but cannot be cited directly as commercial-product evidence.
4. **The drone open-set comparison is not a direct A/B on identical data,** but a comparison across two different datasets of the same genre (classical=DroneRF, deep learning=CardRF). A complete controlled comparison would require rerunning both methods on the same dataset.
5. **Channel estimation has no confirmed result beyond an early experiment with a known methodology flaw.** A corrected redesign exists in code but had not completed execution as of this paper’s writing.
6. **The RF fingerprint K=50 fusion number was flagged in a post-hoc audit for possible cross-session correlation.** The possibility that the 50 observations are correlated, consecutive packets rather than 50 independent observations has not been fully ruled out.

7 Data and Reproducibility

The public datasets cited in this paper are:

- **RadioML2016.10a** - the standard modulation-classification benchmark (O’Shea & West). 6dB SNR subset, 11 classes.
- **DroneRF** - Al-Sa’d et al., Qatar University, Mendeley DOI 10.17632/f4c2b4n755, **CC BY 4.0** (commercially citable with attribution).
- **WiSig** - a public multi-receiver dataset for RF fingerprint research, **CC BY-NC-SA** (non-commercial research use only).
- **INRIA dataset** - used for independent generalization checks in RF fingerprinting, **CC BY** (commercially citable).
- **CardRF** - a drone-controller RF dataset, research-use only (license unverified; a self-captured dataset is recommended for commercial citation).

Every experiment was designed to be reproducible under a fixed seed, and every verdict was decided by a code-computed metric (accuracy, AUROC, macro-F1, SI-SDR, realized false-alarm rate) rather than a self-report from a human or a model. Experiments where a methodology flaw was later found (the first channel-estimation pass, an early interference-detection smoke test) are recorded together with the nature of that flaw, so the same mistake is not repeated.

8 References

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